

deliver the secondary output of 9V, 300 mA. The transformer output is rectified by a full-wave bridge rectifier comprising diodes D1 through D4, filtered by capacitor C1 and then regulated by IC 7805 (IC1). Capacitor C2 bypasses the ripples present in the regulated 5V power supply. LED1 acts as the power-'on' indicator and resistor R1 limits the current through LED1.

An actual-size, single-side PCB for the *pranayam* timer circuit is shown in Fig. 2 and its component layout in Fig. 3. Wire the circuit on the PCB and check for proper connections using a multimeter.

Software

The source program is written in Assembly language and assembled using Metalink's ASM51 assembler, which is freely available on the Internet for download. The source program has been well commented for easy understanding. Burn the 'hex' file code into

PARTS LIST

Semiconductors:

IC1	- 7805, 5V regulator
IC2	- AT89C2051 microcontroller
T1	- SL100 npn transistor
D1-D4	- 1N4007 rectifier diode
LED1	- 5mm LED
LCD	- 16×2 line LCD module

Resistors (all ¼-watt, ±5% carbon):

R1	- 330-ohm
R2-R6	- 10-kilo-ohm
R7	- 2.2-kilo-ohm
R8	- 150-ohm
VR1	- 10-kilo-ohm preset

Capacitors:

C1	- 1000µF, 25V electrolytic
C2, C3	- 0.1µF ceramic disk
C4, C5	- 22pF ceramic disk
C6	- 10µF, 16V electrolytic

Miscellaneous:

X1	- 230V primary to 9V, 300mA secondary transformer
PZ1	- Piezobuzzer
X _{AL}	- 12MHz crystal
S1	- On/off switch
S2-S6	- Push-to-on switch

the microcontroller using a suitable programmer.

The initial count values for *purak*, *kumbhak* and *rechak* are stored in

memory locations 40H, 41H and 42H, respectively. The count values stored in different memory locations are converted into BCD and then into ASCII format to show on the LCD.

Once the countdown for each process reaches '1', the buzzer connected to port pin 3.7 via transistor T1 beeps. When the user presses switch S3, external interrupt 0 generates and the program jumps to vector location 0003H, where the instruction is written to set the count values for *purak*, *kumbhak* and *rechak* in the corresponding memory locations. Switches S4, S5 and S6 are used to set the count values in their corresponding memory locations. The program repeats the countdown process for *purak*, *kumbhak* and *rechak* continuously.

EFY note. The source code of this article has been included in this month's EFY-CD.

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PRANAYAM.ASM

Pranayam.ASM	setb EX1	setb BUZZER
SMOD51	setb IT1	acall l_delay
RS EQU P3.0	setb EX0	clr BUZZER
EN EQU P3.1	setb IT0	mov b,40h
_DATA EQU P1	CLR C	mov a,#0c0h
BUZZER EQU P3.7	MOV 40h,#4 ;Initial timing for	acall command ;Force cursor at
SETTIME EQU P3.2		beginning of the second line
SW1 EQU P3.3 ;SET TIME FOR	PURAK MOV 41h,#16 ;Initial timing for	mov dptr,#msg2
	KUMBHAK MOV 42h,#8 ;Initial timing for	acall disp_msg
	RECHAK MOV P1,#00h ;Configure	d1_loop: mov a,#0c9h
	Port 1 as output port	acall command ;Force cursor at
	mov a,#3bh ;Initialize LCD, 2	beginning of the second line
	lines, 5X7 matrix	acall disp_ss
	acall command ;Issue command to	acall l_delay
	the LCD	DJNZ b,d1_loop
	mov a,#0eh	setb BUZZER
	acall command ;LCD ON, Cursor	lcall l_delay
	ON	clr BUZZER
	mov a,#01h	mov b,41h
	acall command ;Clear LCD	mov a,#0c0h
	mov a,#06h	acall command ;Force cursor at
	acall command ;Shift cursor	beginning of the second line
	right	mov dptr,#msg3
	control_loop: mov a,#01h	acall disp_msg
	acall command ;Clear LCD	d2_loop: mov a,#0c9h
	acall s_delay	acall command ;Force cursor at
	mov a,#80h	beginning of the second line
	acall command ;Force cursor at	acall disp_ss
	beginning of the first line	acall l_delay
	mov dptr,#msg1 ;Point to ini-	DJNZ b,d2_loop
	tial message	setb BUZZER
	acall disp_msg	acall l_delay
		clr BUZZER
		mov b,42h
START: MOV SP,#07h		
setb EA		